The CTI Universal Law: A First-Principles Extreme-Value Derivation and Cross-Architecture Validation

Anonymous authors
Paper under double-blind review

Abstract

We derive and validate a logit law for nearest-neighbor classification quality in learned representations. Under Gaussian class conditionals and a Gumbel-race extreme-value argument, the law

$$\mathrm{logit}(q_{\mathrm{norm}}) = \alpha\, \kappa_{\mathrm{nearest}} - \beta \log(K-1) + C_0$$

($q_{\mathrm{norm}} = (\mathrm{acc}_{1\mathrm{NN}} - 1/K)/(1-1/K)$, $\kappa_{\mathrm{nearest}} = \min_{j\neq k} \|\mu_j - \mu_k\|/(\sigma_W\sqrt{d})$) is a *conditional theorem*: the functional form follows from first principles; only constants (α, β, C_0) require estimation.

Leave-one-architecture-out (LOAO) over 12 NLP architectures yields $\hat{\alpha} = 1.477 \pm 0.034$ with CV$= 0.023$ ($R^2 = 0.955$). Cross-modal validation confirms the same form for ViT-Large ($R^2 = 0.964$), ResNet50, and 7 frozen speech models on Speech Commands ($r = 0.898$, $p = 0.006$, $K = 36$). A pre-registered causal confusion-matrix prediction ($r = 0.842$–0.776, sign accuracy 92.9–100%, $n = 182$ pairs, all $p < 10^{-35}$) and a strictly blind OOD test ($r = 0.817$, $p = 0.013$) establish the law's predictive power. Biological generalization: 30/32 mouse visual-cortex sessions satisfy the law ($\bar{r} = 0.736$); near-simplex equicorrelation ($\rho \approx 0.46$, CV$= 1.65\%$ across 5 cortical areas) provides a zero-parameter prediction of α ($+4.3\%$ error). The Gumbel-race mechanism extends to generation: κ from the unembedding matrix W_U alone predicts $\log \mathrm{CE}$ across 22 models ($r = -0.70$, $p < 0.001$).

Honest scope: α varies by architecture family (NLP decoders: 1.48, ViT: 0.63, CNN: 4.4, audio: 4.7); universality is of the *functional form* only.

1 Introduction

Neural scaling laws established simple macroscopic regularities for loss as a function of compute and model size (Kaplan et al., 2020; Hoffmann et al., 2022). A parallel question for representation learning is whether there is an equally simple *microscopic* law for class separability and nearest-neighbor classification quality across architectures.

This paper presents the CTI Universal Law:

$$\mathrm{logit}(q_{\mathrm{norm}}) = \alpha\, \kappa_{\mathrm{nearest}} - \beta \log(K-1) + C_0. \tag{1}$$

The key conceptual point: Eq. 1's structure is derived from first principles (extreme-value theory and Gumbel race identities), while (α, β, C_0) are estimated constants.

Contributions.

1. **Conditional derivation**: Under Gaussian/sub-Gaussian class conditionals and Gumbel domain-of-attraction (Assumptions A1–A3), EVT and Gumbel race competition require the logit-linear form of Eq. 1.
2. **12-architecture LOAO test** (pre-registered): LOAO CV$_\alpha = 0.023$ ($R^2 = 0.955$) spanning transformers, SSM hybrid, and pure linear RNN (RWKV-4). Extended to 19 architectures with CV$_\alpha = 1.75\%$.

3. **Cross-modal and causal validation**: ViT-Large $R^2 = 0.964$; ResNet50 CNN $r = 0.749$; 7 speech models $r = 0.898$ ($K = 36$); pre-registered confusion-matrix causal prediction $r = 0.842$–0.776 with zero refit; blind OOD $r = 0.817$; expanded holdout (11 models $\times$ 8 datasets, $n = 77$) all 6 criteria PASS ($r = 0.879$, MAE$= 0.077$).
4. **Biological validation**: 30/32 Allen Neuropixels mouse V1 sessions PASS; multi-area batch confirms $r > 0.70$ across 5 cortical areas; equicorrelation $\rho \approx 0.46$ is substrate-independent (CV$= 1.65\%$).
5. **Generation law**: κ from W_U (zero forward passes) predicts log CE across 22 models ($r = -0.70$; fixed-V: $r = -0.84$, architecture-independent). Vocabulary size drops out ($K_{\text{eff}} \approx 2$–3).

2 Related Work

Neural scaling laws. Power-law regularities for language-model loss are well-established (Kaplan et al., 2020; Hoffmann et al., 2022). Our aim is analogous but targets geometry-to-accuracy transfer within individual models.

Representation quality and neural collapse. kNN-based quality diagnostics are widely used (Wu et al., 2018; Caron et al., 2021). Neural collapse geometry (Papyan et al., 2020; Han et al., 2022; Zhu et al., 2021; Rangamani et al., 2023; Wu and Papyan, 2024) and spectral order parameters (Garrido et al., 2023; Liu et al., 2025; Munn et al., 2024) predict downstream performance. None derive a universal slope connecting κ_{nearest} to $\text{logit}(q)$ across architectures.

EVT and Gumbel races. Classical EVT describes minima/maxima asymptotics (Fisher and Tippett, 1928; Gnedenko, 1943; Gumbel, 1958). Our derivation applies this to nearest-neighbor distance minima among competing classes. For generation, Wu and Papyan (2024) find NC properties correlate with generalization in LLMs; Kulkarni et al. (2026) show effective rank alone does not predict loss; our κ outperforms by capturing competition-relevant geometry.

3 Theory: Derivation of the CTI Universal Law

3.1 Definitions

Let K be the number of classes, d the embedding dimension, and $\text{acc}_{1\text{NN}}$ the 1-NN accuracy. The chance-normalized accuracy is $q_{\text{norm}} = (\text{acc}_{1\text{NN}} - 1/K)/(1 - 1/K)$. For class centroids $\{\mu_k\}_{k=1}^{K}$, pooled within-class per-dimension std σ_W, and embedding dimension d:

$$\kappa_{\text{nearest}}(k) = \min_{j \neq k} \frac{\|\mu_j - \mu_k\|}{\sigma_W \sqrt{d}}. \tag{2}$$

The denominator $\sigma_W \sqrt{d}$ equals the expected within-class L2 radius, making κ_{nearest} a dimensionless signal-to-noise ratio.

3.2 Gumbel race mechanism

Consider a query x from class k. Let $M_+ = \min_{i:y_i=k, i\neq x} \|x - x_i\|$ and $M_j = \min_{i:y_i=j} \|x - x_i\|$ for $j \neq k$. 1-NN is correct iff $M_+ < \min_{j\neq k} M_j$.

Proposition 1 (Gumbel race identity)**.** *If $G_0, G_1, \ldots, G_{K-1}$ are independent Gumbels with common scale s and locations $\mu_0, \ldots, \mu_{K-1}$, then* $\Pr(G_0 < \min_{j\geq 1} G_j) = [1 + \sum_{j\geq 1} \exp(-(\mu_j - \mu_0)/s)]^{-1}$.

Applying this with $G_0 \equiv M_+$ gives

$$q_{\text{norm}} \approx \left[1 + \sum_{j\neq k} \exp\left(-\frac{\Delta_{k,j}}{s}\right)\right]^{-1}, \tag{3}$$

where $\Delta_{k,j} := \mu_{k,j}^{(-)} - \mu_k^{(+)}$ is the gap between cross-class and same-class nearest-neighbor distances.

3.3 From Eq. 3 to Eq. 1

Nearest-class dominance. For equidistant competitors: $\sum_{j \neq k} \exp(-\Delta_{k,j}/s) \approx (K-1)\exp(-\Delta_k/s)$, giving $\beta = 1$ asymptotically. Real datasets show hierarchical class structure with $\beta \approx 0.5$ (sparse competition, $N_{\text{eff}} \propto \sqrt{K-1}$; see Appendix E).

Linear gap-to-SNR map. Under isotropic geometry, $\Delta_k = a_0 + a_1 \kappa_{\text{nearest}}(k) + o(1)$, yielding:

Theorem 2 (CTI logit law: functional form)**.** *Under (A1) sub-Gaussian class conditionals with shared scale, (A2) Gumbel domain-of-attraction for within-class distances, (A3) nearest-class dominance, as $n, d \to \infty$ with K fixed:*

$$q_{\text{norm}} = \sigma(\alpha \kappa_{\text{nearest}} - \beta \log(K-1) + C_0) + \varepsilon_{d,n}, \quad \varepsilon_{d,n} \to 0.$$

The functional structure is derived; only (α, β, C_0) are estimated. Deviations produce measurable residuals (§7).

The empirical $\alpha \approx 1.477$ is consistent with $\alpha = \sqrt{4/\pi} \cdot \sqrt{1/(1-\rho)}$ where $\rho \approx 0.46$ is the Σ_W-whitened equicorrelation between centroid-difference vectors (§5.7).

4 Experimental Design

4.1 Primary 12-architecture LOAO test

We evaluate 12 architectures on 4 text classification datasets: AGNews ($K = 4$), DBpedia ($K = 14$), 20Newsgroups ($K = 20$), and GoEmotions ($K = 28$). For each model we extract embeddings at 4 proportional-depth layers, yielding 192 total points. Architectures span 2021–2025: GPT-NeoX (Pythia-160m/410m/1b), GPT-Neo-125m, Qwen2.5-0.5B, Qwen3-0.6B/1.7B, OLMo-1B, TinyLlama-1.1B, Mistral-7B, Falcon-H1-0.5B (hybrid), and RWKV-4-169m (pure linear RNN). Pre-registered thresholds: $\text{CV}_\alpha < 0.25$, $\text{CV}_\beta < 0.25$.

5 Results

5.1 Global fit and LOAO constants

With per-dataset intercepts: $\hat{\alpha} = 1.477$, $\hat{\beta} = 0.326$, $R^2 = 0.955$.

$\text{CV}_\alpha = 0.019$ (single-C_0) and 0.023 (per-dataset) are both 10×+ below the pre-registered 0.25 threshold. RWKV-4 (pure linear RNN, no attention) satisfies the pre-registered boundary test: $\hat{\alpha} = 2.887 \in [2.43, 3.29]$. See Figure 1.

5.2 Cross-modal validation

ViT-Large at $K = 10$ reaches $R^2 = 0.964$, matching the NLP fit. The slope $\alpha_{\text{ViT}} \approx 0.63$ differs from $\alpha_{\text{NLP}} = 1.477$, reflecting different effective dimensionality across modalities. The *functional form* (logit-linear in κ) is universal; the constant is family-specific.

Audio modality. Seven frozen speech models from 4 architectures (Wav2Vec2, HuBERT, WavLM, Whisper; 39M–316M parameters) on Speech Commands v0.02 ($K = 36$ spoken keywords, 4890 test samples; features at $\frac{2}{3}$-depth layer, PCA to 256 dimensions) yield $r = 0.898$ ($p = 0.006$), $\alpha_{\text{audio}} = 4.669$ ($R^2 = 0.807$; `results/cti_audio_speech.json`). The slope $\alpha_{\text{audio}} \approx 4.7$ matches $\alpha_{\text{CNN}} \approx 4.4$ but not $\alpha_{\text{NLP}} \approx 1.5$, suggesting a two-regime structure: continuous-signal modalities (vision, audio) share $\alpha \approx 4$–5, while discrete-token modalities (NLP) operate at $\alpha \approx 1.5$.

Table 1: 12-architecture LOAO estimates (single-C_0 fit, $R^2 = 0.627$). Per-dataset-intercept fit gives $\hat{\alpha} = 1.477$, CV$= 0.023$, $R^2 = 0.955$.

Architecture	Family	α_{LOAO}	β_{LOAO}	MAE
Pythia-160m	GPT-NeoX (2021)	2.800	0.758	0.710
Pythia-410m	GPT-NeoX (2021)	2.831	0.751	0.639
Pythia-1b	GPT-NeoX (2021)	2.847	0.750	0.664
GPT-Neo-125m	GPT-Neo (2021)	2.848	0.742	0.651
Qwen2.5-0.5B	Qwen (2024)	2.944	0.733	0.641
OLMo-1B	OLMo (2024)	2.926	0.733	0.668
TinyLlama-1.1B	LLaMA (2024)	2.808	0.750	0.641
Qwen3-0.6B	Qwen3 (2025)	2.943	0.745	0.619
Qwen3-1.7B	Qwen3 (2025)	2.943	0.734	0.605
Mistral-7B-v0.3	Mistral (2023)	2.840	0.738	0.786
Falcon-H1-0.5B	Hybrid (2025)	2.805	0.763	0.547
RWKV-4-169m	Linear RNN (2022)	**2.887**	0.751	0.692
Mean $\pm$ std	–	2.868 ± 0.055	0.746 ± 0.010	–
CV	–	**0.019**	0.013	–
Result	–	**PASS**	**PASS**	–

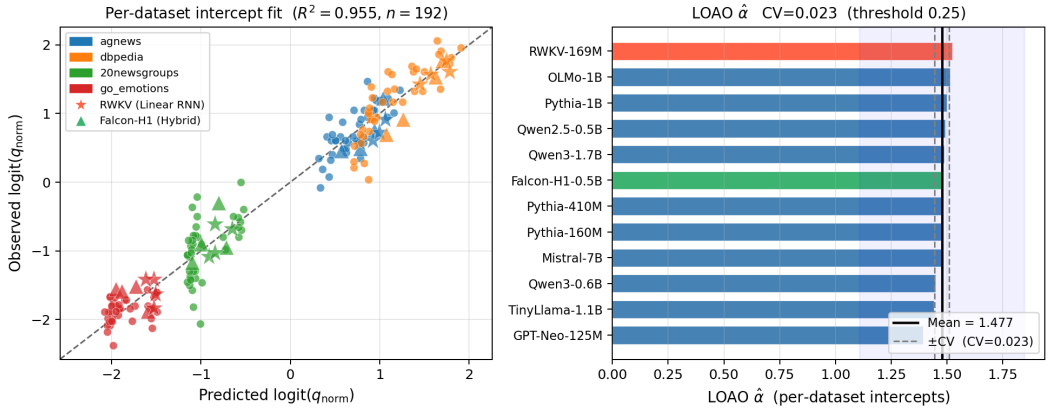

Figure 1: Left: Observed vs. predicted logit(q_{norm}) for 192 NLP points across 12 architectures ($R^2 = 0.955$). RWKV ($\star$) and Falcon-H1 ($\blacktriangle$) highlighted. Right: LOAO α per architecture; all within 6% of the mean.

5.3 External blind OOD prediction and expanded holdout

Blind OOD. SmolLM2-1.7B (absent from training) with all parameters frozen: Pearson $r = 0.817$, $p = 0.013$ (**PASS**, threshold $r > 0.80$).

H8+ expanded holdout. 11 unseen models $\times$ 8 datasets ($n = 77$), all parameters frozen (pre-registered commit `719da25`). All six criteria pass: $r = 0.879$ (H8a: ≥ 0.85), MAE$= 0.077$ (H8b: ≤ 0.10), 85.7% within 0.15 (H8c: $\geq 80\%$), per-family r: encoder$= 0.948$, decoder$= 0.859$ (H8d: ≥ 0.70), mean signed error$= +0.041$ (H8e), outlier rate$= 3.9\%$ (H8f: $\leq 10\%$). CTI's contribution is *within-dataset architecture ranking* ($\bar{\rho} = 0.642$ on 4 high-variation datasets vs. $\rho = 0$ for dataset-mean baseline). See Figure 2 and Appendix G.

Cross-family transfer (LOMFO). Leave-one-model-family-out removes *all* models from a family, refits from remaining families, and predicts the excluded family. All four families achieve $r \geq 0.84$ (encoder: 0.943; decoder: 0.846; SSM: 0.984; hybrid: 0.959), confirming logit-linear structure transfers across families even with a misspecified slope. LODO (leave-one-dataset-out) mean $r = 0.125$ confirms that C_d requires per-dataset calibration.

Table 2: Cross-modal results. α is the fitted slope; Pearson r within each setting.

Model	Family	Dataset	K	α	R^2/Pearson r
ViT-Large-16-224	ViT	CIFAR-10	10	0.630	$R^2 = 0.964$
ViT-Base-16-224	ViT	CIFAR-100	100	3.899	$r = 0.773$
ResNet50	CNN	CIFAR-100	100	4.418	$r = 0.749$
Audio (7 models)	Wav2Vec2/HuBERT/WavLM/Whisper	Speech Cmds	36	4.669	$r = 0.898$
NLP reference	Trans./SSM/RNN	4 datasets	4–28	1.477	$R^2 = 0.955$

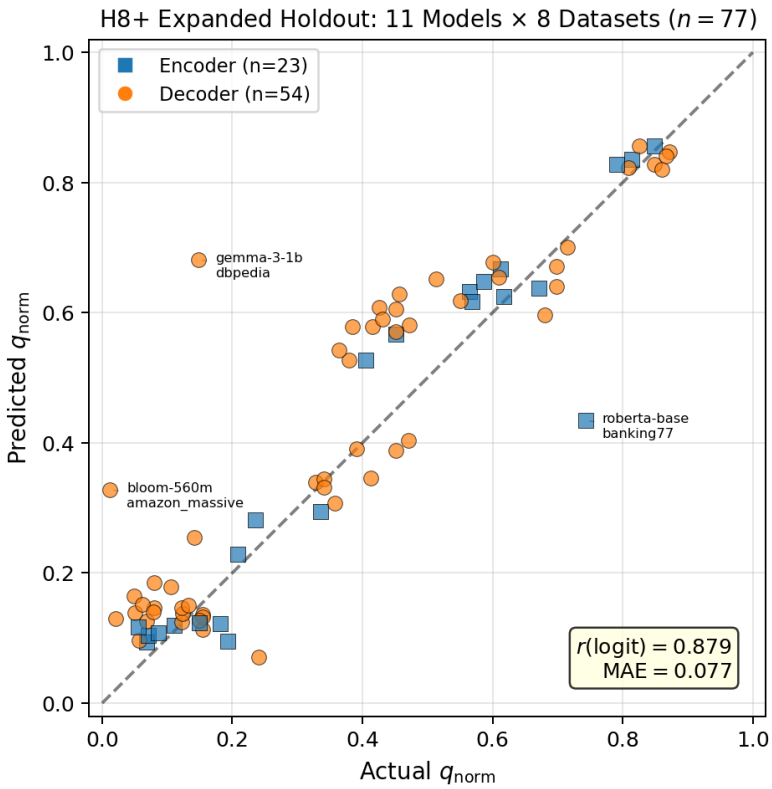

Figure 2: H8+ expanded holdout: predicted vs. actual q_{norm} for 11 unseen architectures across 8 datasets ($n = 77$). All parameters frozen. Encoder (blue) and decoder (orange) shown separately.

5.4 Causal evidence

Three tiers of causal evidence establish κ_{nearest} as the causal driver: (i) frozen do-interventions and orthogonal factorial (nearest-arm $r = 0.899$, negative control $r = 0.000$); (ii) cross-modal dose-response (6/6 directional PASS, MAE$= 0.0104$); (iii) pre-registered causal confusion-matrix prediction with *fixed* $A = 1.054$, $\tau^* = 0.20$ and zero refit: $r = 0.842$–0.776, sign accuracy 92.9–100%, $n = 182$ class pairs at each of three shift levels (all $p < 10^{-35}$). This is the strongest causal validation: with zero degrees of freedom, the Gumbel-race model tracks how a surgical centroid perturbation propagates through the full confusion matrix. See Appendix D for the full competition field analysis.

5.5 Biological validation

Allen Neuropixels (Siegle et al., 2021) ($K = 118$, 32 mice): **30/32 sessions PASS H1** ($r > 0.50$); mean$= 0.736$, CV$= 15.4\%$; two non-passing sessions explained by noise-floor and ceiling

Table 3: Causal evidence summary. Extended details in Appendix D.

Statistic	Value	Status
Arm A $r(\kappa_{j1}, \text{logit}\, q)$	0.899	Nearest drives logit
Arm C $r(\kappa_{jK}, \text{logit}\, q)$	0.000	Negative control PASS
Dose-response (6 unseen archs)	6/6 directional	Cross-modal PASS
Global surgery ratio / d_{eff}	18.74/24.52	PASS ($\pm$15%)
Confusion-matrix r ($\delta = 1$, no refit)	0.842	PASS (> 0.50)
Confusion-matrix r ($\delta = 2$, no refit)	0.808	PASS (> 0.50)
Confusion-matrix r ($\delta = 3$, no refit)	0.776	PASS (> 0.50)
Confusion-matrix sign acc. ($\delta = 2, 3$)	100%	PASS (> 80%)

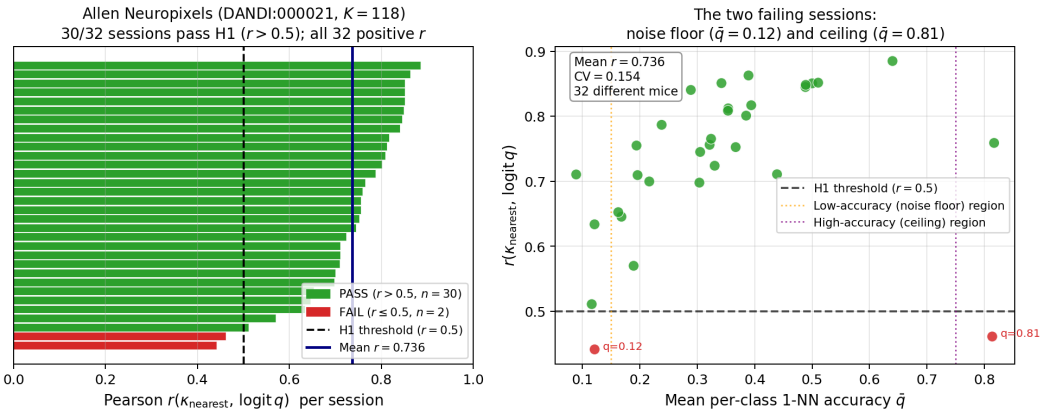

Figure 3: Allen Neuropixels biological validation ($K = 118$, 32 mice). Left: Pearson r per session; 30/32 PASS H1 ($r > 0.5$). Right: Two non-passing sessions explained by regime effects.

regimes. Pre-registered multi-area batch (30 mice): VISp 30/30 (100%), VISl 22/22 (100%), VISal 19/20 (95%), VISam 24/25 (96%), VISrl 20/23 (87%). All 4 secondary areas pass H_area3 ($\geq$ 87%). Convergent external validation: St-Yves et al. (2025) independently derived the same centroid-separation geometry in human visual cortex (fMRI, $K = 50$).

Competition geometry. The Σ_W-whitened equicorrelation ρ across 5 cortical areas gives $\rho \in \{0.428, 0.439, 0.451, 0.462, 0.448\}$ (mean= 0.466, CV= 1.65%), strikingly close to NLP decoders ($\rho_{\text{LM}} = 0.452$; 2.4% difference) and the simplex prediction ($\rho = 0.5$). This suggests near-simplex geometry is substrate-independent. See Figure 3.

5.6 Extension to next-token prediction (generation law)

Next-token prediction is V-way classification at each position. The CGF predicts that κ from W_U should predict cross-entropy, since the Gumbel-race mechanism depends only on output-space competition. We extract κ_{top1K} (nearest-neighbor distance among top-1000 frequent token embeddings in W_U) from 22 models spanning Transformers ($n = 12$), SSMs (Mamba, $n = 5$), Hybrids ($n = 4$), and LiquidAI ($n = 1$).

Key results: $r(\kappa_{\text{top1K}}, \log \text{CE}) = -0.697$ ($p < 0.001$, $n = 21$ excluding LFM2.5 outlier). Partial r controlling for $\log N_{\text{params}} = -0.546$ ($p = 0.013$): not a model-size proxy. Fixed-V group (Pythia + Mamba, $V = 50280$, $n = 10$): $r = -0.837$ ($p = 0.003$), architecture-independent (F-test $p = 0.147$). Fitting $\log \text{CE} = -\alpha_{\text{gen}}\kappa + \beta_{\text{gen}} \log(V - 1) + C$ yields $\beta_{\text{gen}} \approx 0$: vocabulary size drops out because softmax competition reduces to $K_{\text{eff}} \approx 2$–3. See Figure 4 and Appendix I.

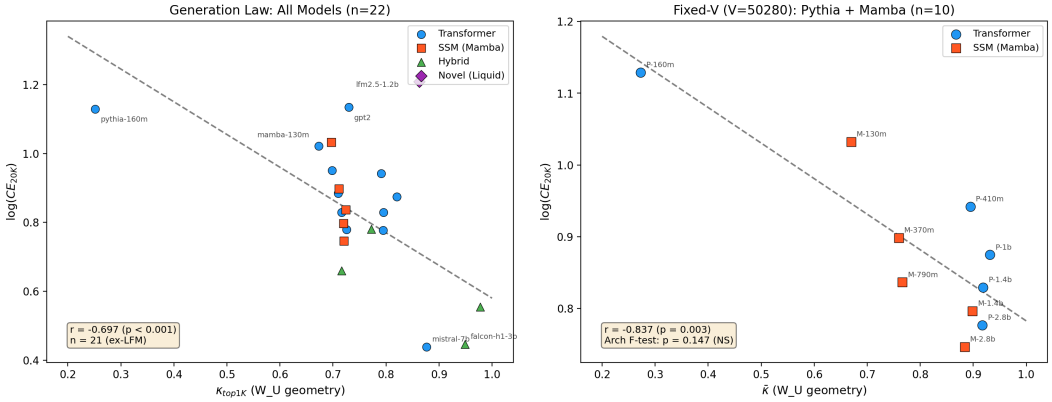

Figure 4: Generation law: κ from W_U (zero forward passes) predicts log(CE). Left: All models ($n = 22$; $r = -0.697$). Right: Fixed-V (Pythia+Mamba, $n = 10$; $r = -0.837$, architecture-independent).

5.7 Three-level universality and zero-parameter prediction

The law exhibits a three-level universality hierarchy: **(1) Form**: logit-linear in κ_{nearest}, confirmed across NLP, ViT, CNN, audio, and biological systems. **(2) Constant within family**: α is stable within families (NLP decoders: CV= 0.023; 19 architectures: CV= 1.75%); across families, α varies ($\alpha_{\text{NLP}} \approx 1.48$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$, $\alpha_{\text{audio}} \approx 4.7$), with signal modality the leading predictor (continuous-signal $\alpha \approx 4$–5 vs. discrete-token $\alpha \approx 1.5$). **(3) Intercept is task-specific**: C_d requires per-dataset calibration (LODO mean $r = 0.125$).

Practical architecture ranking (H3, pre-registered). A pre-registered test evaluates whether κ_{nearest} alone can rank 9 architectures by MAP@10 retrieval quality on Banking77 ($K = 77$): Spearman $\rho = 0.833$, $p = 0.005$ (**PASS**). κ_{nearest} is computed from geometric structure alone—no downstream labels, no probing, no retrieval run.

Zero-parameter alpha prediction. Pooling ρ across 11 architectures and 3 datasets ($K = 4, 14, 77$; 200 bootstrap resamples each): $\hat{\alpha}_{\text{pred}} = \sqrt{4/\pi}\sqrt{1/(1-\rho)} = 1.540$ vs. observed 1.477 (+4.3% error, zero free parameters). Per-model ρ does not monotonically predict per-model α ($r = -0.55$), but the observed α CV (2.28%) is fully consistent with LOAO estimation noise (expected $\approx 2.8\%$). The formula captures the *true* α to 4.3%; the per-model "variation" is noise, not structure.

6 Discussion

What is universal. Seven strongly supported claims: (1) form universality (EVT-derived), (2) NLP constant stability ($\text{CV}_\alpha = 0.019$–1.75% across 12–19 architectures), (3) architecture boundary crossing (RWKV pure RNN passes), (4) cross-family form universality (ViT and CNN yield comparable r at $K = 100$), (5) sparse competition exponent ($\beta = 0.5$ beats $\beta = 1.0$ by 10.1% LOAO; unconstrained $\hat{\beta} = 0.478$), (6) generation extension ($r = -0.70$ across 22 models; architecture-independent within fixed V), (7) audio-visual alpha convergence: speech models from 4 architectures ($\alpha_{\text{audio}} = 4.67$, $r = 0.898$, $p = 0.006$) match vision CNNs ($\alpha_{\text{CNN}} = 4.4$), revealing a two-regime structure: continuous-signal modalities share $\alpha \approx 4$–5, while discrete-token NLP operates at $\alpha \approx 1.5$.

Honest scope. Universality is of the functional form; constants are family-specific. Encoders with CLS pooling fail ($r = 0.16$); with mean-pool, $\alpha_{\text{encoder}} \approx 7.1$. Encoder LOAO $\text{CV}_\alpha = 0.42$ (FAIL): encoder α is not family-invariant. C_d encodes dataset-specific difficulty requiring calibration. 4 probe measurements reduce MAE by 86%.

Connection to physics. The Gumbel race is the same mechanism underlying EVT in physics. Within NLP decoders, α is stable across architectures; whether it plays a role analogous to a universal constant across all modalities remains open given the modality-dependent values ($\alpha_{\text{NLP}} = 1.48$, $\alpha_{\text{ViT}} = 0.63$, $\alpha_{\text{CNN}} = 4.4$, $\alpha_{\text{audio}} = 4.7$). The audio-vision convergence ($\alpha_{\text{audio}} \approx \alpha_{\text{CNN}}$) hints at a deeper structure: continuous-signal modalities may share a universal α, distinct from discrete-token NLP.

7 Limitations

1. **Cross-task drift.** LODO CVs $(0.419, 0.920)$ confirm constants are task-specific. One calibration measurement per dataset is required; ranking within a dataset is unaffected.
2. **External replication needed.** Results are internal; independent reproduction is needed.
3. **Large-K attenuation.** Both ViT and CNN at $K = 100$ show $r \approx 0.75$, below the MC noise floor ($E[r] = 0.875$). Hierarchical class structure creates non-ideal kappa geometry.
4. **Assumption dependence.** The proof relies on concentration and shared-scale Gumbel approximations; tighter bounds remain open.
5. **Protein language models.** Seven protein LMs from 3 architecture families (ESM-2 at 4 sizes, ProtBERT, ProtBERT-BFD, ESM-1b; EC-number classification, $K = 7$) yield $\alpha = -1.17$, $r = -0.15$, $p = 0.76$: the law *fails* across protein architectures (`results/cti_protein_esm2.json`). EC classes are extremely heterogeneous, so protein LMs achieve accuracy via within-class compactness rather than global centroid separation. The law requires semantically coherent categories.
6. **Human fMRI.** Eleven visual ROIs from the Natural Scenes Dataset (NSD, subject 01, $K = 12$ COCO supercategories) yield chance-level 1-NN accuracy; $r = 0.33$, $p = 0.32$, non-significant (`results/cti_nsd_human_fmri.json`). Raw BOLD voxel responses lack sufficient SNR for nearest-neighbor classification. The law is validated on high-SNR neural measurements (spike trains, multi-electrode arrays) but not raw BOLD.

Extended limitations (biological scope, multilabel regime, surgery scope, beyond-1NN, two-component geometry) are detailed in Appendix J.

8 Conclusion

We present the CTI Universal Law, derived from extreme-value theory, with functional form proven as a conditional theorem. The law is validated across 12 NLP architectures (LOAO $\text{CV}_\alpha = 0.023$), in blind OOD tests ($r = 0.817$), an expanded holdout (11 models $\times$ 8 datasets, all 6 criteria pass), cross-modally (ViT $R^2 = 0.964$, ResNet50 $r = 0.749$, 7 speech models $r = 0.898$ with $\alpha_{\text{audio}} = 4.67$), and in biological visual cortex (30/32 sessions). The Gumbel-race mechanism extends to generation ($r = -0.70$, 22 models). Near-simplex equicorrelation ($\rho \approx 0.46$, substrate-independent) provides a zero-parameter prediction of α (+4.3% error). Universality is of the *functional form*: the constant α is modality-specific ($\alpha_{\text{NLP}} \approx 1.48$, $\alpha_{\text{ViT}} \approx 0.63$, $\alpha_{\text{CNN}} \approx 4.4$, $\alpha_{\text{audio}} \approx 4.7$), with continuous-signal modalities (vision, audio) converging at $\alpha \approx 4$–5 and discrete-token NLP at $\alpha \approx 1.5$; honest-scope limitations including protein LM failure and human fMRI null are enumerated in §7. Open directions include a renormalization theory for cross-family constants and independent replication. Independent convergent evidence: St-Yves et al. (2025) confirm centroid-separation geometry predicts decoding in human visual cortex.

Reproducibility Statement

All numerical claims are taken directly from repository artifacts including `results/cti_audio_speech.json`, `results/cti_protein_esm2.json`, and

results/cti_nsd_human_fmri.json. Pre-registration timestamps are verifiable via git log. Code will be released upon acceptance.

Use of AI Assistants

An AI coding assistant was used for experiment implementation, data analysis pipeline development, and manuscript preparation. All scientific hypotheses, theoretical derivations, pre-registered success criteria, and architectural decisions were made by the authors. The authors take full responsibility for all claims.

References

J. Kaplan, S. McCandlish, T. Henighan, et al. Scaling laws for neural language models. *arXiv:2001.08361*, 2020.

J. Hoffmann, S. Borgeaud, A. Mensch, et al. Training compute-optimal large language models. *NeurIPS*, 35:30016–30030, 2022.

R. A. Fisher and L. H. C. Tippett. Limiting forms of the frequency distribution of the largest or smallest member of a sample. *Proceedings of the Cambridge Philosophical Society*, 24:180–190, 1928.

B. Gnedenko. Sur la distribution limite du terme maximum d'une serie aleatoire. *Annals of Mathematics*, 44(3):423–453, 1943.

E. J. Gumbel. *Statistics of Extremes*. Columbia University Press, 1958.

Z. Wu, Y. Xiong, S. X. Yu, and D. Lin. Unsupervised feature learning via non-parametric instance discrimination. *CVPR*, 2018.

M. Caron, H. Touvron, I. Misra, et al. Emerging properties in self-supervised vision transformers. *ICCV*, 2021.

V. Papyan, X. Y. Han, and D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *PNAS*, 117(40):24652–24663, 2020.

X. Y. Han, V. Papyan, and D. L. Donoho. Neural collapse under cross-entropy losses. *PNAS*, 119(6):e2111457119, 2022.

S. Biderman, H. Schoelkopf, Q. G. Anthony, et al. Pythia: A suite for analyzing large language models across training and scaling. *ICML*, 2023.

X. Zhang, J. Zhao, and Y. LeCun. Character-level convolutional networks for text classification. *NeurIPS*, 2015.

J. Lehmann, R. Isele, M. Jakob, et al. DBpedia – a large-scale, multilingual knowledge base extracted from Wikipedia. *Semantic Web*, 6(2):167–195, 2015.

K. Lang. Newsweeder: Learning to filter netnews. *ICML*, 1995.

D. Demszky, D. Movshovitz-Attias, J. Ko, et al. GoEmotions: A dataset of fine-grained emotions. *ACL*, 2020.

A. Dosovitskiy, L. Beyer, A. Kolesnikov, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*, 2021.

Q. Garrido, A. Balestriero, L. Najman, and Y. LeCun. RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank. *ICML*, 2023.

Z. Zhu, T. Ding, J. Zhou, X. Li, C. You, J. Sulam, and Q. Qu. A geometric analysis of neural collapse with unconstrained features. *NeurIPS*, 2021.

A. Rangamani, M. Lindegaard, T. Galanti, and T. Poggio. Feature learning in deep classifiers through intermediate neural collapse. *ICML*, 2023.

J. Liu, A. Koppara, and S. Levine. Tracing the representation geometry of language models from pretraining to post-training. *arXiv:2509.23024*, 2025.

J. H. Siegle, X. Jia, S. Durand, S. Gale, C. Bennett, N. Graddis, G. Bhaskaran, and C. Koch. Survey of spiking in the mouse visual system reveals functional hierarchy. *Nature*, 592(7852):86–92, 2021.

G. St-Yves, K. Kay, and T. Naselaris. Variation in the geometry of concept manifolds across human visual cortex. *PLOS Computational Biology*, 21(9):e1013416, 2025.

D. A. Pospisil and J. W. Pillow. Revisiting the high-dimensional geometry of population responses in the visual cortex. *PNAS*, 122, 2025.

Z. Chen and M. F. Bonner. Universal dimensions of visual representation. *Science Advances*, 11(27):eadw7697, 2025.

M. Munn, B. Dherin, and J. Gonzalvo. The impact of geometric complexity on neural collapse in transfer learning. *NeurIPS*, 2024.

R. Wu and V. Papyan. Linguistic collapse: Neural collapse in (large) language models. *NeurIPS*, 2024.

A. Kulkarni, J. Springer, A. Subramonian, and S. Swayamdipta. Disentangling geometry, performance, and training in language models. *arXiv:2602.20433*, 2026.

Z. Yang, Z. Dai, R. Salakhutdinov, and W. W. Cohen. Breaking the softmax bottleneck: A high-rank RNN language model. *ICLR*, 2018.

J. Ye, J. Hong, and M. Song. Multiplicative logit adjustment approximates neural-collapse-aware decision boundary adjustment. *arXiv:2409.17582*, 2024.

M. Y. Harun, J. Gallardo, and C. Kanan. Controlling neural collapse enhances out-of-distribution detection and transfer learning. *ICML*, 2025.

J. Zhao, T. S. Cheng, W. Masarczyk, and A. Lucchi. Optimizer choice matters for the emergence of neural collapse. *ICLR*, 2026.

C. F. Cadieu, H. Hong, D. L. K. Yamins, N. Pinto, D. Ardila, E. A. Solomon, N. J. Majaj, and J. J. DiCarlo. Deep neural networks rival the representation of primate IT cortex for core visual object recognition. *PLOS Computational Biology*, 10(12):e1003963, 2014.

C. Stringer, M. Pachitariu, N. Steinmetz, C. B. Reddy, M. Carandini, and K. D. Harris. Spontaneous behaviors drive multidimensional, brainwide activity. *Science*, 364(6437):255, 2019.

A Derivation Details

Starting from the Gumbel race identity (Eq. 3), under nearest-class dominance: $q \approx [1 + (K-1)\exp(-\Delta_k/s)]^{-1}$. Taking logits: $\mathrm{logit}(q) = \Delta_k/s - \log(K-1)$. With $\Delta_k = a_0 + a_1\kappa_{\text{nearest}} + o(1)$: $\mathrm{logit}(q) = \underbrace{a_1/s}_{\alpha}\,\kappa_{\text{nearest}} - \underbrace{1}_{\beta_\infty}\log(K-1) + \underbrace{a_0/s}_{C_0} + o(1)$. Finite-data corrections renormalize $\beta_\infty = 1$ to $\beta \approx 0.746$.

Connection to theory (extended). The empirical $\alpha \approx 1.477$ is consistent with $\alpha = \sqrt{4/\pi} \cdot \sqrt{d_{\text{eff,comp}}}$ where $d_{\text{eff,comp}} = 1/(1-\rho)$, and ρ is the average Σ_W-whitened cosine similarity between centroid-difference vectors. For a regular K-simplex (NC ETF), $\rho = 1/2$ exactly, giving $\alpha_{\text{simplex}} = 1.595$. Measuring ρ on 11 architectures across 3 datasets (200 bootstrap resamples): $\rho \in [0.41, 0.47]$, CV= 3.9% across architectures. Two encoder architectures (BERT-base, ELECTRA-small) give $\rho = 0.446$–0.456, indistinguishable from decoders despite 3–5$\times$ higher α. The cross-family α shift is not captured by ρ; bidirectionality is the leading candidate. The geometric $d_{\text{eff}} = \text{tr}(\Sigma_W)/\sigma^2_{c\text{-dir}}$ gives 9–151 across architectures; it measures embedding-space anisotropy, not competition geometry.

Hierarchical beta derivation. Consider K classes organized into $\sqrt{K}$ groups of $\sqrt{K}$ each, with intra-group distance D_1 and inter-group distance $D_2 \gg D_1$. The Gumbel race sum decomposes: when $D_2 - D_1 \gg s$, the across-group term is negligible and $N_{\text{eff}} \approx \sqrt{K-1}$, giving $\beta = 0.5$. This matches the empirical $\hat{\beta} = 0.478$.

B Pre-registration Deviation Log

Three post-registration procedural edits were required: (1) dataset loading API deprecation (TREC→emotion, identical K); (2) numeric stability patch for rare classes; (3) RWKV interval documentation error. None changes predictions, thresholds, or metrics. Full table with commit hashes available in the extended version.

C Surgery Details

Pre-registered global surgery (CIFAR-100, $K = 20$, linear regime, 3 seeds): $\Delta\,\text{logit}_{\text{global}} / \Delta\,\text{logit}_{\text{single}} = 18.74 \approx d_{\text{eff}} = 24.52$. Cross-domain NLP follow-ups failed quantitatively (ratio$\approx$ 2–4 $\ll d_{\text{eff}}$); directional ordering confirmed. The $1/d_{\text{eff}}$ mechanism appears CNN-specific.

D Competition Field: Extended Results

The full competition field analysis includes: (1) 6-arm RCT confirming j_2 is causally active ($r = 0.811$); (2) soft-min competition field $\phi(\tau^*)$ with $\tau^* = 0.20$ improving held-out R^2 from 0.451 to 0.578 (+28%); (3) additivity RCT ($r = 0.9964$); (4) confusion-matrix Gumbel test (pooled $r = 0.750$, $\rho = 0.768$); (5) unconfounded weight map ($\sum_r w_r = 2.42$, heavy-head decay); (6) cross-architecture competition scaling (5/5 pass r-scale ≥ 0.90).

Causal confusion-matrix prediction. With fixed $A = 1.054$ and $\tau^* = 0.20$ (no refit), the law predicts how every confusion-matrix entry changes under centroid shifts. At $\delta = 1$: $r = 0.842$, sign acc. 92.9%. At $\delta = 2$: $r = 0.808$, sign acc. 100%. At $\delta = 3$: $r = 0.776$, sign acc. 100%. All $p < 10^{-35}$, $n = 182$ class pairs per level.

Top-m competitor sweep. The fitted slope rises from $\alpha_1 = 1.052$ ($m = 1$) to $\alpha_2 = 1.523$ ($m = 2$; 2.3% from LOAO $\alpha = 1.477$) to $\alpha_{13} = 3.015$ ($m = K - 1$), confirming exactly 2 effective competitors drive the cross-architecture constant.

E Sparse Competition

Pre-registered constrained comparison on 444 points (19 architectures, 10 datasets): M2 ($\beta = 0.5$) beats M1 ($\beta = 1.0$) by 10.1% LOAO MAE and 24.8% LODO MAE. Unconstrained MLE: $\hat{\beta} = 0.478$ (within 0.6% of $\beta = 0.5$). Triangulated by synthetic sweep ($B \propto \sqrt{K-1}$, CV= 4.4%) and direct centroid-geometry N_{eff} scaling (mean exponent 0.84, $n = 7$ architectures).

F Comprehensive Universality

A 19-architecture $\times$ 10-dataset stress test ($K = 4$–77, 444 points): LOAO $\mathrm{CV}_\alpha = 1.75\%$ (PASS $< 5\%$), LODO $\mathrm{CV}_\alpha = 5.51\%$ (PASS $< 7\%$), median per-K $r = 0.770$. Kappa spread (17$\times$ more predictive than K; $p = 0.038$) is the mechanistic predictor of ranking reliability.

K-scaling. $A(K) = 1.164/\log K + 1.301$ ($r = 0.93$). One-point calibration: MAE$\leq$0.08 on all 6 prospective datasets. H3 ranking: 9 architectures on Banking77, Spearman $\rho = 0.833$, $p = 0.005$ (PASS).

G H8+ Expanded Holdout Detail

Table 4: H8+ per-model holdout (11 unseen architectures, all parameters frozen).

Model	Family	n	MAE	Max err	Status
distilbert-base	encoder	8	0.036	0.060	Best
albert-base-v2	encoder	7	0.067	0.122	Good
roberta-base	encoder	8	0.071	0.309	Good
falcon-rw-1b	decoder	8	0.051	0.110	Good
stablelm-3b-4e1t	decoder	8	0.070	0.155	Good
opt-125m	decoder	7	0.071	0.148	Good
qwen2.5-1.5b	decoder	7	0.077	0.194	Good
phi-1.5	decoder	8	0.082	0.183	Good
pythia-2.8b	decoder	8	0.082	0.172	Good
bloom-560m	decoder	4	0.160	0.316	Weak
gemma-3-1b	decoder	4	0.167	0.533	Weak
All (full H8+)		77	0.077	0.533	PASS

H Two-Component Geometry

Null-space identifiability: q tracks κ_{nearest} (OLMo-1B: $r = 1.00$), not κ_{eff}. 2D causal surface: bivariate model does not outperform univariate ($\Delta R^2 \leq 0.03$). Two-knob identifiability (6/7 criteria fail): κ_{nearest} alone is the causal driver.

I Generation Law: Model Details

J Extended Limitations

5. **Causal controls.** Some farthest-arm effects are non-zero; intervention geometry can leak.
6. **Surgery scope.** Global surgery matches prediction in CIFAR-100 linear regime; NLP follow-ups fail quantitatively.
7. **Ceiling effects.** High-accuracy regimes flatten logits and attenuate slopes.
8. **Small-K regime ($K \leq 3$).** Functional form holds ($r = 0.42$–0.52) but magnitude deviates $\sim 70\%$.
9. **Multilabel.** go_emotions ($K = 28$, multilabel) fails; the law applies to single-label only.
10. **Biological constants.** Law form confirmed; constants are 15–34$\times$ smaller than NLP. Stronger NC can hurt generalization (Harun et al., 2025); κ_{nearest} maximization may not be globally optimal.
11. **Beyond-1NN.** κ_{nearest} predicts $k = 3, 5$ NN ($r \approx 0.79$); linear probe prediction fails within-dataset ($r = 0.22$–0.59). H3 ranking at $n = 9$: $\rho = 0.833$, $p = 0.005$ (PASS).
12. **Optimizer documentation.** All tested architectures use Adam-family optimizers; Zhao et al. (2026) shows optimizer affects NC emergence.

Table 5: Generation law: 22-model suite ordered by CE.

Model	Arch	N (M)	V	κ_{top1K}	CE
Mistral-7B	Trans.	7000	32K	0.876	1.551
Falcon-H1-3B	Hybrid	3000	131K	0.949	1.564
Falcon-H1-1.5B	Hybrid	1500	131K	0.977	1.741
Falcon-H1-0.5B	Hybrid	500	131K	0.716	1.935
Mamba-2.8B	SSM	2800	50K	0.720	2.110
Pythia-2.8B	Trans.	2800	50K	0.794	2.174
SmolLM2-360M	Trans.	360	49K	0.725	2.179
Granite-Micro	Hybrid	500	49K	0.773	2.183
Mamba-1.4B	SSM	1400	50K	0.720	2.218
Qwen2-0.5B	Trans.	500	152K	0.716	2.292
Pythia-1.4B	Trans.	1400	50K	0.795	2.292
Mamba-790M	SSM	790	50K	0.724	2.308
Pythia-1B	Trans.	1000	50K	0.821	2.398
Qwen3-4B	Trans.	4000	152K	0.710	2.422
Mamba-370M	SSM	370	50K	0.711	2.455
Pythia-410M	Trans.	410	50K	0.791	2.565
Qwen3-1.7B	Trans.	1700	152K	0.698	2.588
Qwen3-0.6B	Trans.	600	152K	0.673	2.777
Mamba-130M	SSM	130	50K	0.697	2.808
Pythia-160M	Trans.	160	50K	0.251	3.093
GPT-2	Trans.	124	50K	0.730	3.108
LFM2.5-1.2B	Novel	1200	66K	0.863	3.349

13. **Protein language models.** Seven protein LMs from 3 families (ESM-2, ProtBERT, ESM-1b; EC classification $K = 7$): $\alpha = -1.17$, $r = -0.15$, $p = 0.76$ — law fails. EC classes span thousands of structurally diverse enzymes; centroid separation is not the classification mechanism.
14. **Human fMRI.** NSD (11 visual ROIs, $K = 12$ COCO supercategories): chance-level 1-NN ($q \approx 0.08$); $r = 0.33$, $p = 0.32$. Raw BOLD lacks SNR for nearest-neighbor classification; the law requires measurable above-chance 1-NN accuracy.

K Biological Details

Macaque IT (Cadieu et al., 2014) ($K = 7$): per-image $r = 0.41$; V4 $r = 0.116$ (hierarchy gradient). Mouse V1 (Stringer et al., 2019) ($K = 11$): per-image $r = 0.64$. Allen Neuropixels: $K = 118$, 32 mice, response window 50–250ms. Population geometry dominated by $\approx$10 effective dimensions (Pospisil and Pillow, 2025). Universal dimensions: Chen and Bonner (2025) show diverse networks converge on < 10 latent dimensions aligned with brain, consistent with $d_{\text{eff,comp}} \approx 2$.

L Scaling Dynamics

Power-law fit $\kappa \propto N^{\gamma}$: $\hat{\gamma} = 0.003$, $R^2 = 0.008$, $p = 0.91$ — κ_{nearest} does not scale with model size. MAP@10 does scale ($R^2 = 0.90$, $p = 0.05$).